

Frayed Demazure weaves for Poisson-compatible cluster structures on Bott–Samelson charts

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Dramatis Personae

- G : simple algebraic group over $\mathbb{C}$.
- B : Borel subgroup.
- I : index set of simple roots.
- $\varphi_i : SL_2 \hookrightarrow G$ root embedding.
- $\underline{w} = (i_1, i_2, \dots, i_\ell) \in I^\ell$: word
- $\gamma \in \{i_1, -i_1\} \times \dots \times \{i_\ell, -i_\ell\}$: subexpression of $\underline{w}$.

Bott–Samelson varieties

For $i \in I$, $P_i = B \sqcup Bs_iB$ is a minimal parabolic subgroup of G .

Consider the product $P_{\underline{w}} := P_{i_1} \times P_{i_2} \times \cdots \times P_{i_\ell} \subset G^\ell$ with B^ℓ -action

$$(g_1, g_2, \dots, g_\ell) * (b_1, \dots, b_\ell) = (g_1 b_1, b_1^{-1} g_2 b_2, \dots, b_{\ell-1}^{-1} g_\ell b_\ell).$$

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The *Bott–Samelson variety* is the quotient $Z_{\underline{w}} := P_{\underline{w}}/B^\ell$.

Configurations of flags

A B^ℓ -equivariant map $P_{\underline{w}} \hookrightarrow G^\ell$, gives an embedding

$$Z_{\underline{w}} \hookrightarrow (G/B)^\ell, \quad [g_1, g_2, \dots, g_\ell] \mapsto (g_1 B, g_1 g_2 B, \dots, g_1 \cdots g_\ell B).$$

Each point is a sequence of flags where the j -th flag differs from the previous at most in position i_j .

Affine charts

$$B_i(z) := \varphi_i \begin{pmatrix} z & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad B_{-i}(z) := \varphi_i \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix}.$$

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For $\gamma \in \{i_1, -i_1\} \times \{i_2, -i_2\} \times \cdots \times \{i_\ell, -i_\ell\}$ a subexpression of $\underline{w}$, the *Bott–Samelson chart* $\mathcal{O}^\gamma$ is the image of the embedding

$$\Phi_\gamma : \mathbb{A}^\ell \hookrightarrow Z_{\underline{w}}, \quad (z_1, \dots, z_\ell) \mapsto [B_{\gamma_1}(z_1), B_{\gamma_2}(z_2), \dots, B_{\gamma_\ell}(z_\ell)].$$

Standard Poisson structures

- The group G has a *standard Poisson structure*.
This is a Lie bracket $\{-, -\}_G : \mathbb{C}[G] \times \mathbb{C}[G] \rightarrow \mathbb{C}[G]$ which is a $\mathbb{C}$ -derivation in both entries, and respects the group multiplication.

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- Parabolic subgroups P_i are Poisson subvarieties.
- Product Poisson structure on $P_{\underline{w}} = P_{i_1} \times \cdots \times P_{i_\ell}$.
- Pushforward to quotient $Z_{\underline{w}} = P_{\underline{w}}/B^\ell$.

Poisson compatibility

Definition (Gekhtman–Shapiro–Vainshtein)

A cluster structure and a Poisson structure are *compatible* if every (extended) cluster is log-canonical.

The Elek–Lu formula

Theorem (Elek–Lu, '21)

For $j < k$, the Poisson structure on $\mathcal{O}^\gamma \subset Z_{\underline{w}}$ has the form

$$\{z_j, z_k\}_\gamma = \begin{cases} C_{j,k}^\gamma z_j z_k & \text{if } \gamma_j < 0, \end{cases}$$

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Here, $C_{j,k}^\gamma$ are integer constants determined by Cartan integer data, and the polynomial $\sigma_{j,k}^\gamma \in \mathbb{C}[z_{j+1}, \dots, z_k]$ is determined by the action of a particular vector field.

$$\underline{w} = (1, 2, 1, 2) \text{ and } \gamma = (1, 2, 1, -2)$$

Example

In the coordinates for the Bott–Samelson charts $\mathcal{O}^{\underline{w}}$ and $\mathcal{O}^{\gamma}$ respectively:

$$\begin{aligned} \{z_1, z_2\}_{\underline{w}} &= -z_1 z_2, & \{z_1, z_3\}_{\underline{w}} &= z_1 z_3 - 2z_2, & \{z_1, z_4\}_{\underline{w}} &= 2z_1 z_4 - 2, \\ \{z_2, z_3\}_{\underline{w}} &= -z_2 z_3, & \{z_2, z_4\}_{\underline{w}} &= z_2 z_4 - 2z_3, & \{z_3, z_4\}_{\underline{w}} &= -z_3 z_4. \end{aligned}$$

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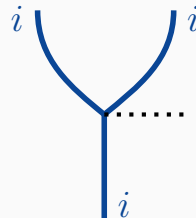
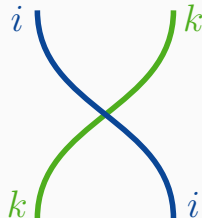
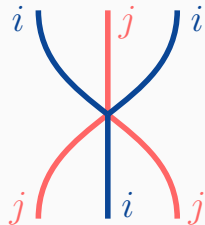
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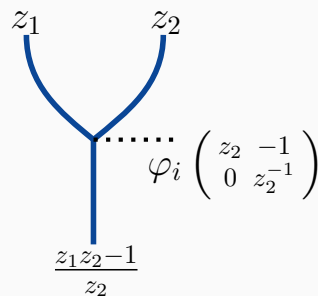
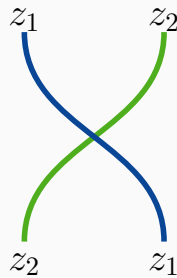
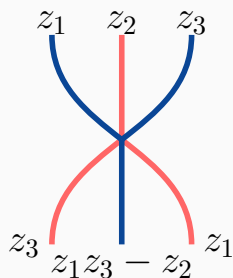
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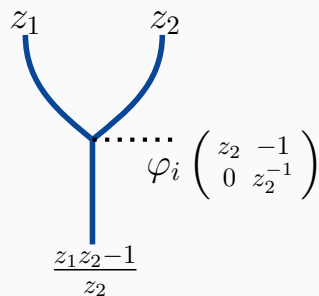
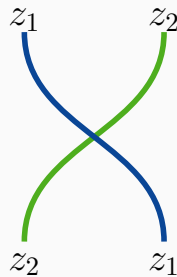
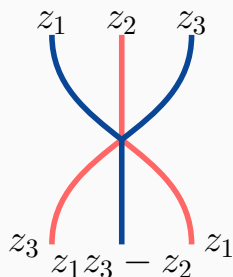
Demazure weaves [CGGS24,CGGLSS25]



Demazure weaves



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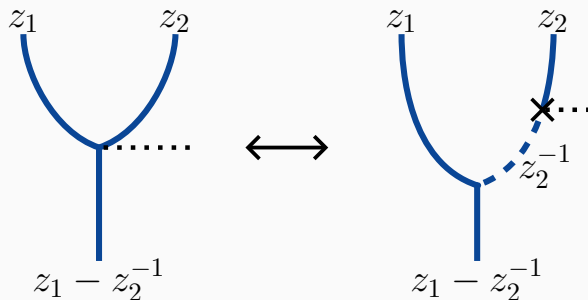
These give Poisson maps between Bott–Samelson cells.

$$f^*(\{z_j, z_k\}_{\underline{w}}) = \{f^*(z_j), f^*(z_k)\}_{\underline{w}'}$$

Easy to compute for the first two.

Poisson-ness of weaves

For the third map, it factors via the change of coordinates to an adjacent Bott–Samelson chart.



Weaves and the standard Poisson structure

Theorem (C. '26)

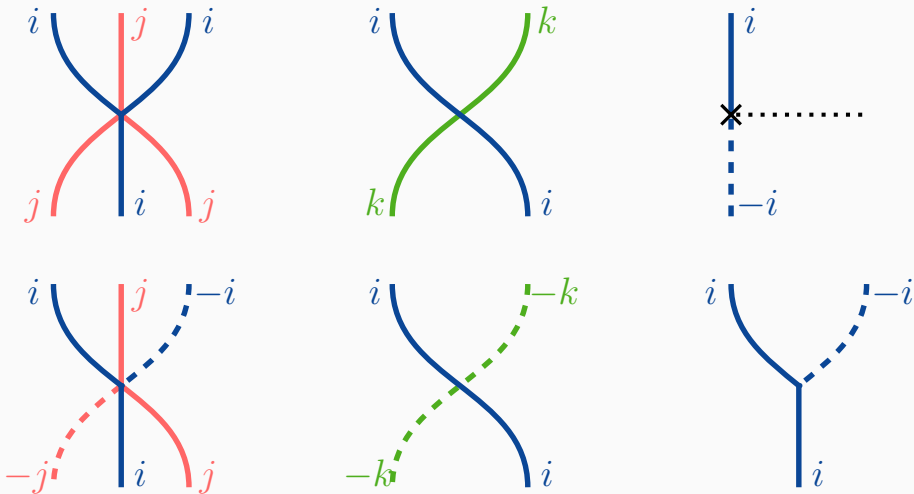
- *Weave tori are log-canonical in the standard Poisson structure.*

Weaves and the standard Poisson structure

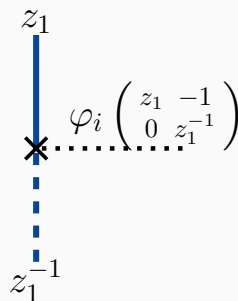
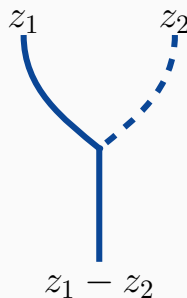
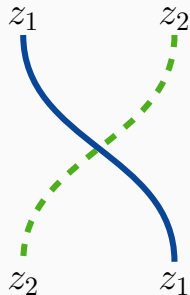
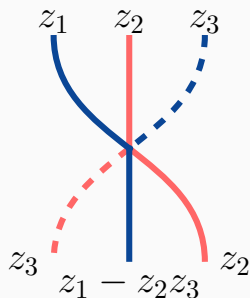
Theorem (C. '26)

- *Weave tori are log-canonical in the standard Poisson structure.*
- *The cluster structures on Bott–Samelson cells and braid varieties are compatible with the standard Poisson structure.*

Frayed strands for weaves

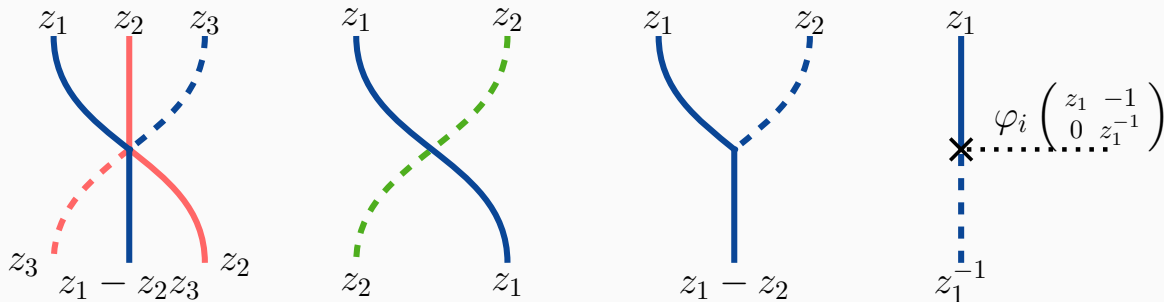


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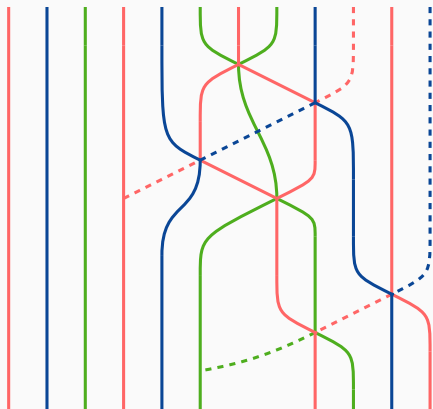


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Proposition

For $(\underline{w}_0; \gamma)$, the all but last are enough to twine all frayed strands.

The map $\rho : \mathcal{O}^\gamma \rightarrow \mathcal{O}^{\gamma^+}$



$$(\underline{w}_0; 2, 3, 1, -2, 2, -1)$$

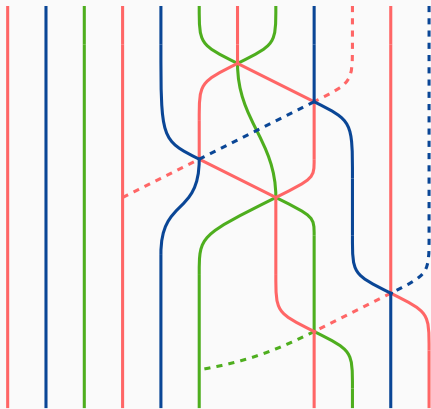
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The map $\rho : \mathcal{O}^\gamma \rightarrow \mathcal{O}^{\gamma^+}$

Theorem (C. '26, also Lu–Yakimov (unpublished))

If γ has m negative letters, we have a Poisson isomorphism

$$\mathcal{O}^\gamma \xrightarrow{\cong} \mathcal{O}^{\gamma^+} \times \mathbb{A}^m,$$

$$\Phi_\gamma(\underline{z}) \mapsto (\rho(\Phi_\gamma(\underline{z})) , (z_j)_{\gamma_j < 0})$$

where the RHS has a log-canonical product structure.

The map $\rho : \mathcal{O}^\gamma \rightarrow \mathcal{O}^{\gamma^+}$

$$\begin{array}{ccccc}
 \mathcal{O}^{\underline{w}} & \xleftarrow{\quad \text{dashed} \quad} & \mathcal{O}^\gamma & \xleftarrow{\quad \cong \quad} & \mathcal{O}^{\gamma^+} \times \mathbb{A}^m \\
 & \searrow \text{dashed} & \downarrow \rho & & \swarrow p_1 \\
 & & \mathcal{O}^{\gamma^+} & &
 \end{array}$$

Cluster structures on $\mathcal{O}^\gamma$

Theorem (C. '26)

- *There is a Poisson-compatible cluster structure on $\mathcal{O}^\gamma$ with seed*

$$\mathbf{s}(\gamma) := \rho^*(\mathbf{s}(\gamma^+)) \sqcup \left\{ \boxed{z_j} \mid \gamma_j < 0 \right\}.$$

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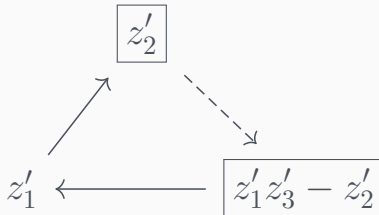
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The sequence $\overrightarrow{\mu}_\gamma$ is defined by negating letters from left to right.

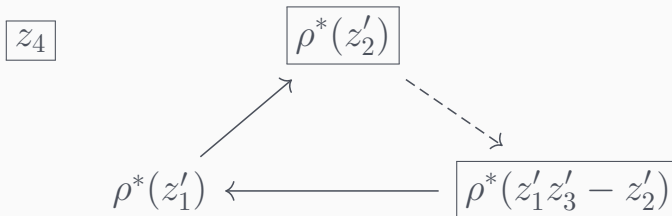
$$\gamma = (1, 2, 1, -2)$$

The seed $s(\gamma^+)$ on $\mathcal{O}^{\gamma^+}$ in the BS coordinates z'_i for $\mathcal{O}^{\gamma^+}$:



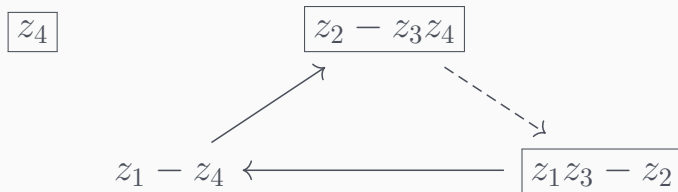
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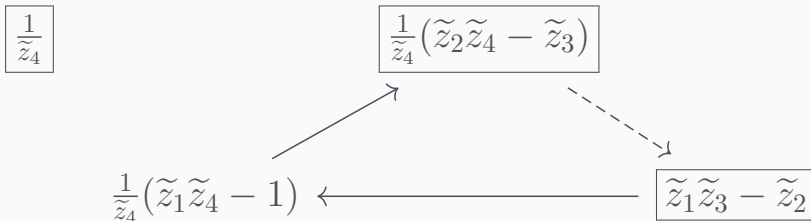
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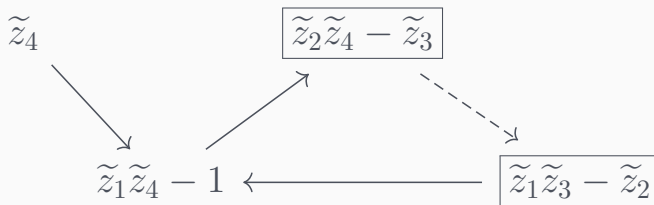
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The seed $s(\gamma)$ in the BS coordinates $\tilde{z}_i$ for $\mathcal{O}^w$:



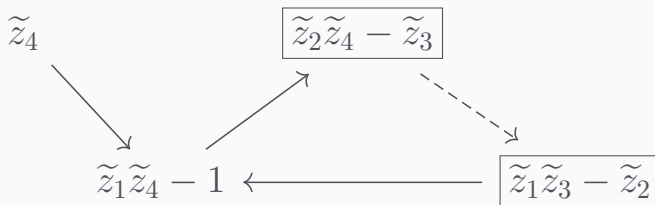
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The seed $\vec{\mu}_\gamma(\mathbf{s}(\underline{w}))$ in the BS coordinates $\tilde{z}_i$ for $\mathcal{O}^w$:

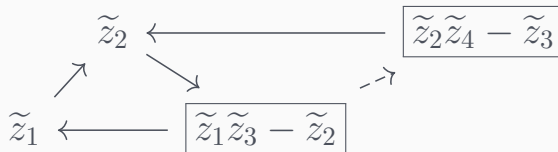


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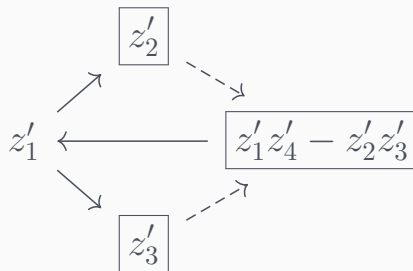


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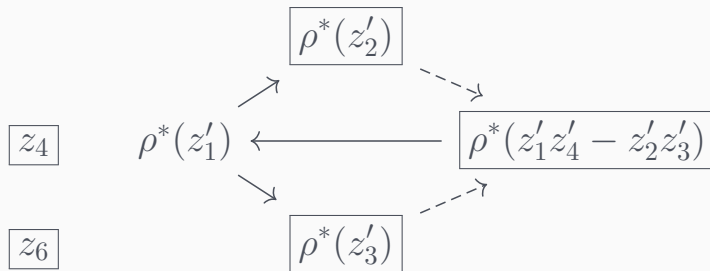


$$\gamma = (2, 3, 1, -2, 2, -1)$$

The seed $s(\gamma^+)$ on $\mathcal{O}^{\gamma^+}$ in the BS coordinates z'_i for $\mathcal{O}^{\gamma^+}$:

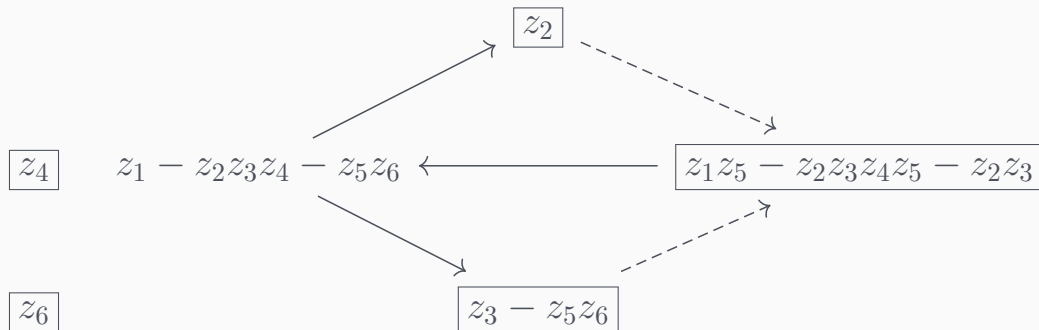


The seed $s(\gamma)$ on $\mathcal{O}^\gamma$:



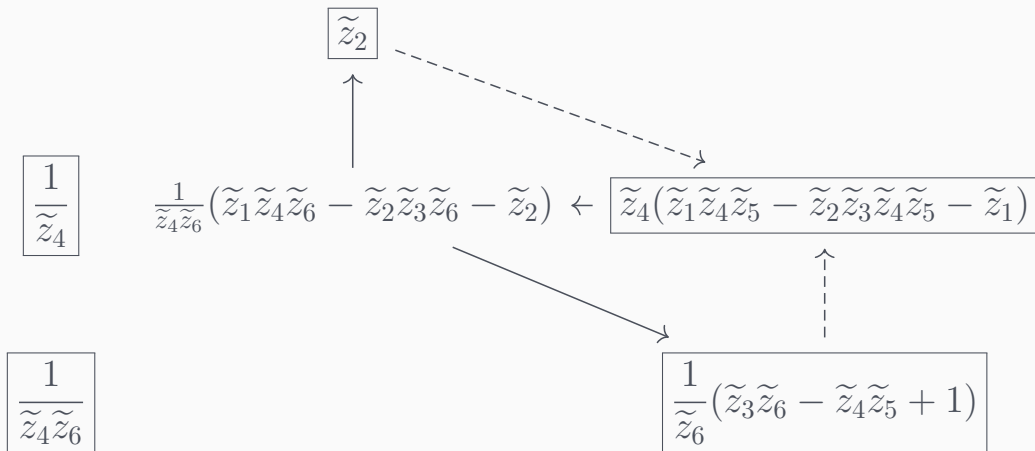
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The seed $s(\gamma)$ in the BS coordinates z_i for $\mathcal{O}^\gamma$:



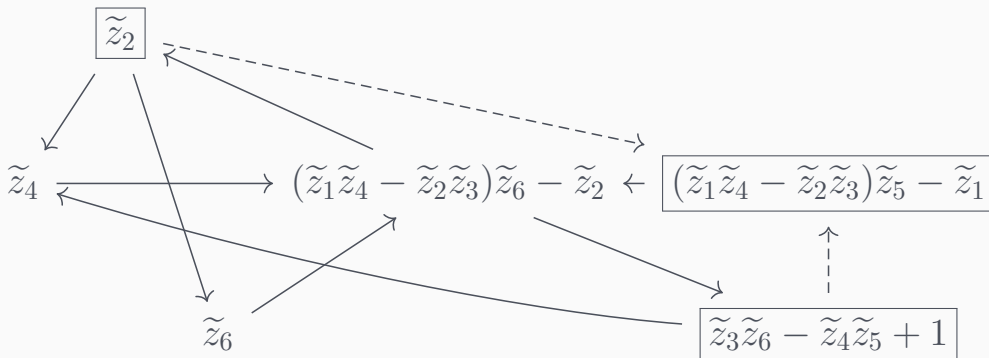
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The seed $s(\gamma)$ in the BS coordinates $\tilde{z}_i$ for $\mathcal{O}^w$:



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The seed $\vec{\mu}_\gamma(s(\underline{w}))$ in the BS coordinates $\tilde{z}_i$ for $\mathcal{O}^{\underline{w}}$:

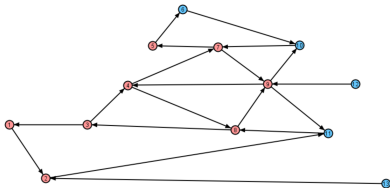


Mutation sequences

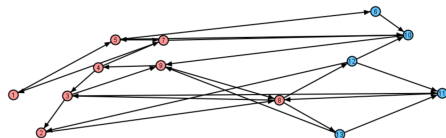
Remark

This seed $\overrightarrow{\mu}_\gamma(s(\underline{w}))$ differs from Ménard's open Richardson seed $M(\gamma)(s(\underline{w}))$ by an application of $\overrightarrow{\mu}_{\gamma^+}$.

$$\gamma = (2, 1, -2, 3, -4, 5, 4, 2, -3, -4, -2, -3, -1)$$



$s(\underline{w})$



$\vec{\mu}_\gamma(s(\underline{w}))$